



The Mismatched Message

The messages don't match — fix the link

Name: _____

Date: _____

★ I can find why a message does not start a script and fix the words so they match.

Pixel never hears it

Bit (green) starts on the green flag and sends a message. Pixel (purple) waits for a message. Pixel is supposed to reach 3 — but it never moves! Look closely: the word Bit SENDS and the word Pixel WAITS FOR are not the same. Find the mismatched message and fix it.

Pixel's goal: reach 3

0 Bit	1	2	3 ★ goal	4	5 Pixel
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Bit's script

when green flag clicked

forward 1

broadcast 'go'

Pixel's script (has a bug)

when I receive 'stop'

back 2

1 Run it — what goes wrong?

Click the flag. Where does Bit land? Does Pixel move at all? Circle YES or NO. What number is Pixel on?

2 Find the mismatch

Write the two messages. Bit broadcasts ' ____ '. Pixel waits for ' ____ '. Are they the same word? Why doesn't Pixel start?

3 Fix it

Change ONE message so they match. Write Pixel's fixed hat (or Bit's fixed broadcast). After the fix, where does Pixel land?

Big idea

A 'when I receive' script only starts if the broadcast message is EXACTLY the word it waits for. Here Bit sent 'go' but Pixel waited for 'stop', so Pixel never started. Make the two messages the same word and Pixel runs to its goal.



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Quick facilitation notes & answer key

What this worksheet teaches

Students debug a broken message link. Pixel never starts because the message it waits for ("stop") does not match the message Bit sends ("go"). Students compare the two words and fix one so they match. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel) so children can trace both sets of moves and see which one misses the goal.

How to lead it (about 20 minutes)

1. Show them (you do). Click the flag: Bit moves, but Pixel stays home — its message never matched.
2. Do one together. Exercise 1 — Bit lands on 1, Pixel does not move. Exercise 2 — Bit sends "go", Pixel waits for "stop".
3. Let them try. Students change one message so they match, then run Pixel (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 0, forward 1 → 1. Pixel does NOT move (NO); it stays on its home number 5.

Exercise 2 — Bit sends "go"; Pixel waits for "stop". They are not the same word, so Pixel's event never happens.

Exercise 3 — fix: change Pixel's hat to "when I receive go" (or Bit's message to "stop"). With matching messages, Pixel runs: 5, back 2 → 3. Either fix works as long as both words match.

If students get stuck — and ways to adjust

Watch for: changing the moves instead of the message. The bug is the mismatched word.

Make it easier: write the two words side by side and make them the same.

Make it harder: ask which other word would work, as long as both match.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, find and fix the wrong one, and explain the fix.

You'll know it worked when

The child makes the two messages match so Pixel runs.